

BCA 3rd SEMESTER EXAM., 2014

JAVA PROGRAMMING CODE- 303303

Time: 3 hours

Full Marks: 60

Instructions:

- i. The Marks are indicated in the right -hand margin.
- ii. There are **SEVEN** questions in this paper.
- iii. Attempts **FIVE** question in all.
- iv. Question Nos. **1** and **2** are compulsory.

1. Choose the correct one / Answer question of the following (any six). 2*6=12

(a) A methods in an abstract class must be declared abstract.

(I) True

(II) False

(b) What is the value of Y in a given program segment?

```
Int a=5;    int b=6;    Y= b++ + ++a;    System.out.println(Y);
```

(c) If m and n are int type variables, what will be the result of the expression m%n,

When m= -14 and n= -3 ?

(I) 4

(II) 2

(III) -2

(IV) None of the above

(d) Which of the following statement is valid array declaration?

(I) int x [];

(II) float y[];

(III) double [] z;

(IV) t int[];

(e) The methods wait () and notify () are defined in

(I) Java. long.string

(II) Java. long.Runnable

(III) Java. long.object

(IV) Java. long.Thread

(V) Java. long.Thread Group

(f) Given the following code:

```
Class ABC {int x=10; }
```

```
Class PQR extends ABC { int x=20; }
```

```
ABC b=new ABC ();
```

```
PQR d=new PQR ();
```

```
ABC bd= new PQR ();
```

The statement -

`System.out.println(b.x+" "+d.x+" "+bd.x);`

Will produce the output

(I) 10 20 20

(II) 10 20 10

(III) 20 10 20

(IV) 20 20 10

(g) The set Background () method is part of the class

(I) Graphics

(II) Applet

(III) Component

(IV) Container

(V) Object

(h) When we implement the runnable interface, we must define the method

(I) start ()

(II) init ()

(III) run ()

(IV) runnable ()

(V) resume ()

(VI) main ()

(i) Which of the following keyword is used to control access to a member?

(I) default

(II) abstract

(III) protected

(IV) interface

(V) public

(j) Which of the following will produce a value of 10, if x=9.7?

(I) floor (x)

(II) abs(x)

(III) rint (x)

(IV) round(x)

(V) ceil ()

2. Answer any three of the following:

4*3=12

(a) How is Java strongly associated with the internet?

(b) What is empty statement? Explain its usefulness?

(c) What is constructor? What are its special properties?

(d) What is vector? How is it different from an array?

(e) How do we tell Java that we want to use a particular package in a file?

Answer any three of the following:

12*3=36

3. What is the significance of 'super' keyword in an inheritance? Explain with an example how

Subclass class can superclass constructor and how super keyword can access superclass member.

4. What is static import? How is it useful? Using package access and importing packages

Technique, write a program to find the largest among three numbers.

5. Describe the complete life cycle of a thread. Write a program to illustrate “implementing the Runnable Interface”

6. Explain how to rethrow an exception. Write a program to create a try block that is likely to generate three types of exception and then incorporate necessary catch blocks to catch and handle them appropriately .

7. Write down the methods defined by Input Stream. Write a Java program that demonstrates random access I/O which writes six doubles to a file and then reads them back in non-sequential order.

Code : 303303

BCA 3rd Semester Theory Examination, 2017

Java Programming

3 Hrs

Full Marks : 60

Instructions :

The questions are of equal value.

1) *There are Seven questions in this Paper.*

i) *Attempt Five questions in all.*

ii) *Question Nos. 1 & 2 are compulsory.*

Answer any six of the following:

2×6=12

List various Literals in Java.

1) Explain Byte code concept of Java.

State the purpose of break and continue statements in java.

Define an applet.

Define an interface.

What is class path?

List the applet classes in java.

List the advantages of exceptions.

List any three methods of Frame class in Java.

List the different awt components in Java.

2. Answer any three of the following:

(a) Explain data type conversion and casting feature with an example.

(b) Explain use of command line arguments.

(c) Explain for loop with an example.

(d) Explain how variables are declared and initialized.

(e) Explain briefly about access modifiers and their use.

Answer any three

3×12=

3. Explain about abstract classes and abstract methods.

4. Explain life cycle of a Thread.

5. Explain about try-catch functionality in Exception Handling.

6. Explain various Stream Classes of Java.

7. Define Inheritance. Give various forms of Inheritance and write a Java program on single inheritance.

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BCA 3rd Semester Exam., 2018

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

GROUP—A

1. Choose the correct answer of the following
(any six) : 2×6=12

(a) Literal can be of which of these data types?

- (i) Integer
- (ii) Float
- (iii) Boolean
- (iv) All of the above

- (b) Which of these cannot be used for a variable name in Java?

- (i) Identifier
- (ii) Keyword
- (iii) Identifier and keyword

- (c) Decrement operator ' - ' decreases value of variable by what number?

- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

- (d) What is the output of relational operators?

- (i) Integer
- (ii) Boolean
- (iii) Characters
- (iv) Double

- (e) Which of these operators is used to allocate memory for an object?

- (i) Malloc
- (ii) Alloc
- (iii) New
- (iv) Give

(f) Which of the following is a method having same name as that of its class?

- (i) Finalize
- (ii) Delete
- (iii) Class
- (iv) Constructor

(g) What will s2 contain after following lines of code?

```
String s1 = "one";  
String s2 = s1.concat("two")
```

- (i) one
- (ii) two
- (iii) onetwo
- (iv) twoone

(h) Which of these keywords is used to define interfaces in Java?

- (i) interface
- (ii) Interface
- (iii) intf
- (iv) Intf

(i) When do exceptions in Java arise in code sequence?

- (i) Run time
- (ii) Compilation time
- (iii) Can occur any time
- (iv) None of the above

(j) Which of these functions is called to display the output of an applet?

- (i) display()
- (ii) print()
- (iii) displayApplet()
- (iv) PrintApplet()

GROUP—B

2. Answer any three of the following :

- (a) How to set up environment for executing a Java Program? Explain.
- (b) What do you mean by scope and lifetime of variables?
- (c) What do you mean by decision making?
- (d) Explain life cycle of a thread.
- (e) Can we overload main() method? Explain.

GROUP - C

Answer any three questions of the following :

12×3=36

3. (a) What are different OOPs concepts?
(b) What is difference between object-oriented programming language and object-based programming language?
4. (a) What do you mean by operator? Discuss logical operator, assignment operator, miscellaneous operators and operators precedence.
(b) Write a Java program to swap the contents of two numbers using Bitwise XOR operation.
5. (a) What are different ways to compare Java String? Explain any two.
(b) Write a program in Java to implement binary search.
6. (a) How to catch exceptions in Java? Explain with suitable example.
(b) Write an applet program to display moving banner.

(Turn Over)

7. (a) What do you mean by reading and writing files in Java?
(b) What do you mean by directories in Java? How to create and list directories. Demonstrate using a program.

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BCA 3rd Semester Exam., 2019

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer (any six) : $2 \times 6 = 12$

(a) The new operator

- (i) returns a pointer to a variable
- (ii) creates a variable called new
- (iii) obtains memory for a new variable
- (iv) tells how much memory is available

(b) Command to execute a compiled Java program is

- (i) javac
- (ii) java
- (iii) run
- (iv) execute

20AK/411

(Turn Over)

(2)

(c) The order of the three top level elements of the Java source file is

- (i) Import, Package, Class
- (ii) Class, Import, Package
- (iii) Package, Import, Class
- (iv) Random order

(d) Which one of the following is not supported by OOP?

- (i) Abstraction
- (ii) Polymorphism
- (iii) Encapsulation
- (iv) Global variables

(e) From the following statements which is a disadvantage of a Java array?

- (i) An array can hold primitive types data.
- (ii) An array has its size that is known as array length.
- (iii) An array knows only its type that it contains. Array type is checked at the compile time.
- (iv) An array holds only one type of data.

20AK/411

(Continue)

- (f) Which method is used to change the name of a thread?
- (i) Public String getName()
 - (ii) Public void setName (String name)
 - (iii) Public void getName()
 - (iv) Public String setName (String name)
- (g) Which is used for reading streams of raw bytes such as image data and for reading streams of characters, consider using FileReader?
- (i) FileInputStream
 - (ii) FileOutputStream
 - (iii) Both (i) and (ii)
 - (iv) None of the above
- (h) An exception is an event, which occurs during the execution of a program that disrupts the normal flow of the program's instructions?
- (i) True
 - (ii) False
- (i) A package can be renamed without renaming the directory in which the classes are stored.
- (i) True
 - (ii) False

(Turn Over)

- (j) Which exception occurs when a class is not found while dynamically loading a class using the class loaders?
- (i) ClassNotFoundException
 - (ii) ClassFoundException
 - (iii) NoClassDefFoundError
 - (iv) ClassDefFoundError
2. Answer any *three* of the following :
- (a) Explain why Java is machine independent.
 - (b) Explain any two ways of 'static' keyword usage.
 - (c) Brief on constructor overloading.
 - (d) Differentiate between extend and implement keywords.
 - (e) Differentiate between abstract class and interface in Java.
3. Explain the concepts of object-oriented programming supported by Java.
4. What is the role of 'finally' in exception handling? Explain with an example.
5. Discuss in detail about Applet life cycle.
6. What is an array? How are arrays declared, object created and arrays are accessed? Explain with examples.
7. Differentiate method overloading with method overriding with examples.

Code : 303303

(2)

BCA 3rd Semester Exam., 2020

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Answer any six questions : 2×6=12

- (a) List out Java keywords.
- (b) Write about one of the SELECTION statements of Java.
- (c) List out the properties of static variables and methods.
- (d) Write about NEW operator.
- (e) List out thread properties.
- (f) List out some of the sources of errors in programming.

(g) What is byte code in Java?

(h) Define an applet.

(i) What is a package?

(j) What is class path?

2. Answer any three of the following : 4×3=12

- (a) Define an array and explain one-dimensional arrays in Java with an example.
- (b) Define a class in Java.
- (c) Explain the use of command-line arguments.
- (d) Explain about deadlock.
- (e) Explain thread priority.

3. Explain the concept of dynamic method dispatch with an example program. 12

4. Explain the following : 12

- (a) Try and catch
- (b) Throw
- (c) Finally

(3)

5. Define the method overloading. Write a program using the concept of method overloading. 12
6. Explain the life cycle of an applet. 12
7. Define inheritance. Give various forms of inheritance and write a Java program on single inheritance. 12

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Code : 303303

BCA 3rd Semester Exam., 2022

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. **1** and **2** are compulsory.*

1. Answer any six of the following : 2×6=12

- (a) Define type conversion.**
- (b) What is the scope of variable?**
- (c) Why is Java called an architectural neutral?**
- (d) What is the purpose of byte stream?**
- (e) What is the use of final keyword?**
- (f) Define class and object.**

AK23/1024

(Turn Over)

- (g) What is the use of package?
- (h) What is inheritance?
- (i) What is the use of try block?
- (j) What is dynamic binding?

2. Answer any *three* of the following : 4×3=12

- (a) Write a program to find the sum of digit of the number entered by user.
- (b) Explain array list and iterator methods of collections.
- (c) Write a Java program to implement runnable interface with example.
- (d) How are threads synchronized?
- (e) Contrast overloading and overriding methods.

3. Explain briefly about string class and discuss various methods in string class with example.

12

4. Discuss the various levels of access protection available for packages and their implications.

12

5. What is an exception? Explain how an exception can be handled in Java. Also list the benefits of exception handling. 12
6. Explain byte stream class in detail. Differentiate between input stream class and reader class. 12
7. Differentiate between applet and application. Explain applet life cycle with suitable diagram. 12

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BCA 3rd Semester Exam., 2024

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following questions :

2×6=12

- (a) What is World Wide Web?
- (b) Which keyword is used to define a class in Java?
- (c) What is the data type for single character in Java?
- (d) Which operator is used for assignment in Java?
- (e) What does JVM stand for?
- (f) What is 'do' statement?

- (g) What is the process of combining data and methods into a single unit called?
- (h) Which is applet?
- (i) What is the superclass of all classes in Java?
- (j) Illustrate the concept of 'nesting of method'.

2. Answer any *three* of the following : 4×3=12

- (a) Explain the difference between an abstract class and an interface in Java.
- (b) What is method overloading in Java? Explain its benefits with a simple example demonstrating how it works.
- (c) Describe the various looping statements of Java programming.
- (d) What are the key principles of Object-Oriented Programming (OOP)? Briefly explain how Java supports at least two of these principles.
- (e) Explain the concept of inheritance in Java.

3. Write a simple Java program. What is the purpose of a Java virtual machine ? 12

4. What are class and object in Java? How is class related to an object? How is an object created from a class? 12
5. How do applets differ from applications? Describe applet life cycle. 12
6. Explain the fundamental difference between byte streams and character streams in Java's I/O system. When would you typically use one over the other? Provide an example of a class representing each type of stream. 12
7. Discuss the following concepts : 3×4=12
- (a) Java token
 - (b) Java API package
 - (c) While statement
 - (d) Else-if ladder

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BCA 3rd Semester Exam., 2025

JAVA PROGRAMMING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six from the following questions :

2×6=12

- (a) What is Java Programming?
- (b) List the various data types in Java.
- (c) What is the data type for single character in Java?
- (d) Which operator is used for assignment in Java?
- (e) What does JVM stand for?
- (f) What is 'do' statement?

- (g) What is the process of combining data and methods into a single unit called?
- (h) What is applet?
- (i) What is the superclass of all classes in Java?
- (j) Illustrate the concept of 'nesting of method'.

2. Answer any *three* from the following questions : 4×3=12

- (a) Explain the difference between an abstract class and an interface in Java.
- (b) What is method overloading in Java? Explain its benefits with a simple example demonstrating how it works.
- (c) Describe the various looping statements of Java Programming.
- (d) What are the key principles of Object-Oriented Programming (OOP)? Briefly explain how Java supports at least two of these principles.
- (e) Explain the concept of inheritance in Java.

(3)

3. Write a simple Java program. What is the purpose of a Java Virtual Machine? 12
4. What are class and object in Java? How is class related to an object? How is an object created from a class? 12
5. How do applets differ from applications? Describe applet life cycle. 12
6. Explain the fundamental difference between byte streams and character streams in Java's I/O system. When would you typically use one over the other? Provide an example of a class representing each type of stream. 12
7. Discuss the following concepts : 3×4=12
 - (a) Java Virtual Machine
 - (b) Java API package
 - (c) While statement
 - (d) Else-if ladder

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